EXPERIMENT 14: ANALYSIS OF EDFA-BASED DENSE WDM LINK PERFORMANCE USING OPTISYSTEM

OBJECTIVES :

1. To design and simulate a Dense Wavelength Division Multiplexing (DWDM) optical communication system using OptiSystem.

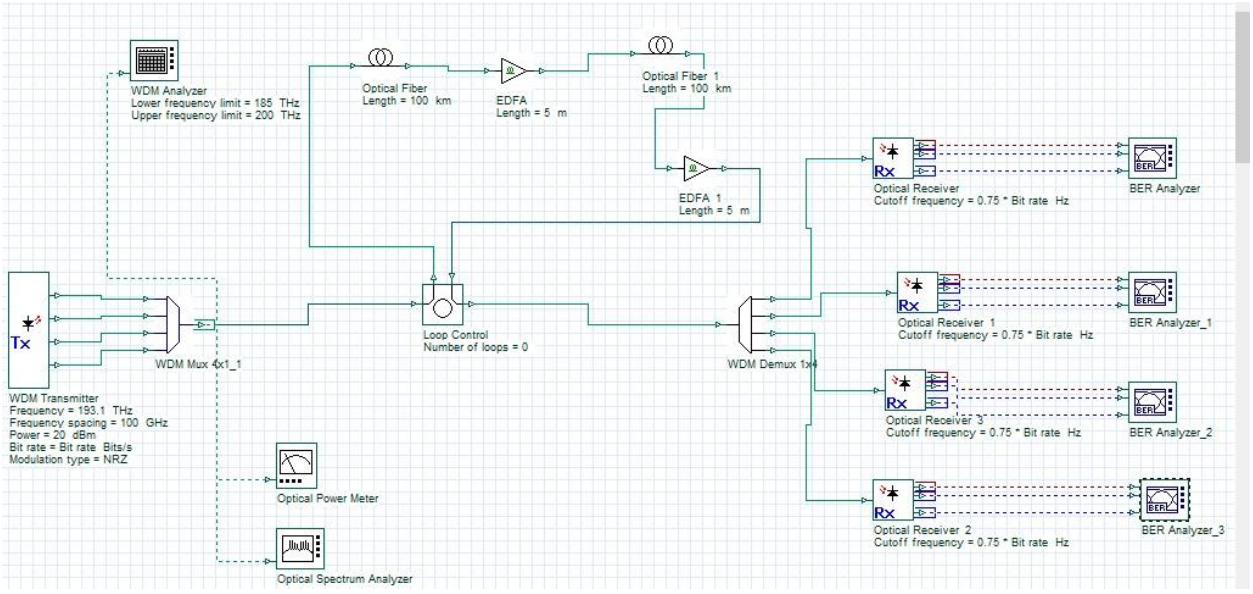
2. To investigate the impact of varying optical fiber length on overall system performance.

3. To examine the effectiveness of an Erbium-Doped Fiber Amplifier (EDFA) in compensating for fiber a enuation losses.

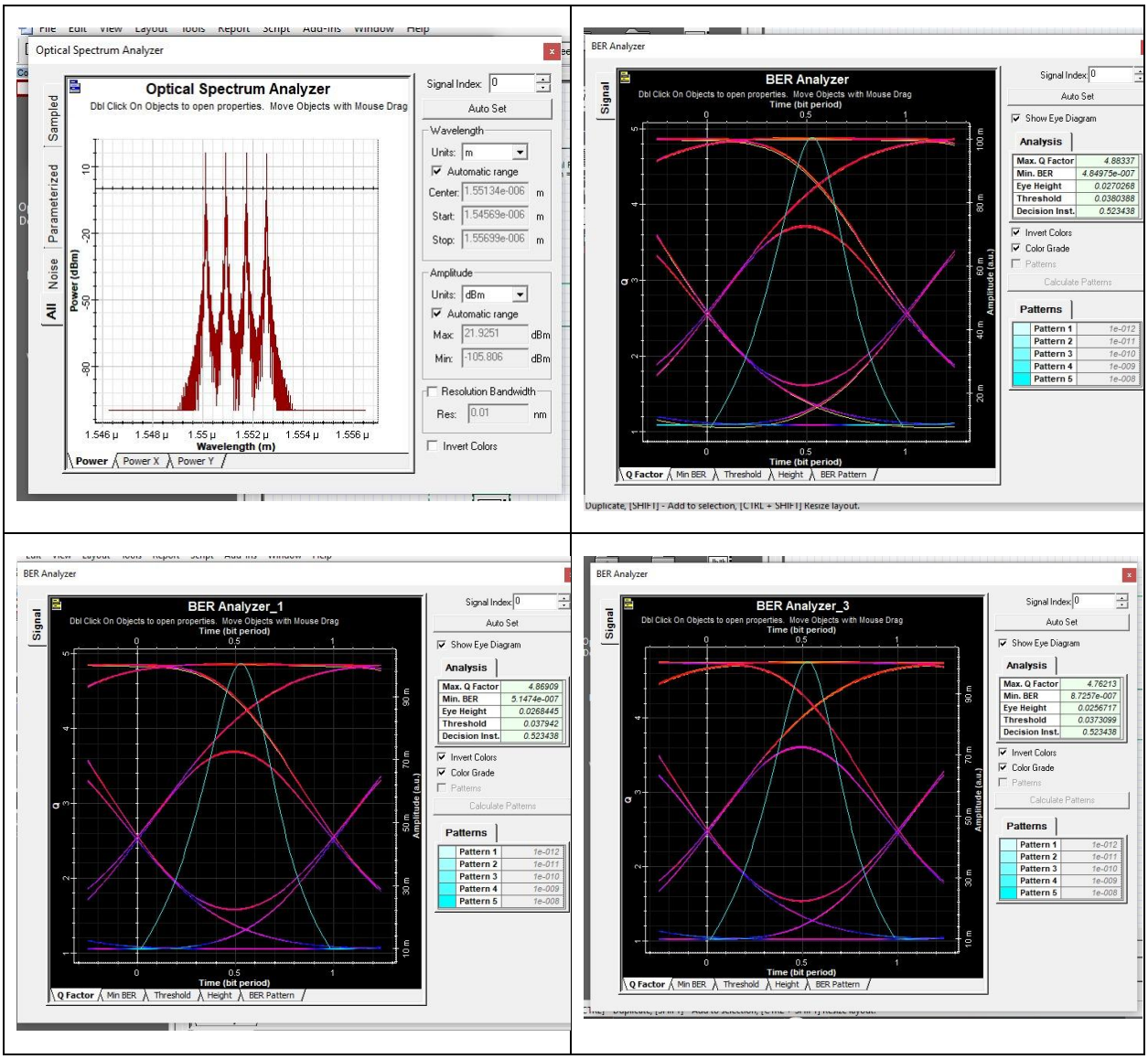
4. To evaluate and compare received power, bit error rate (BER), and Q-factor across different fiber lengths.

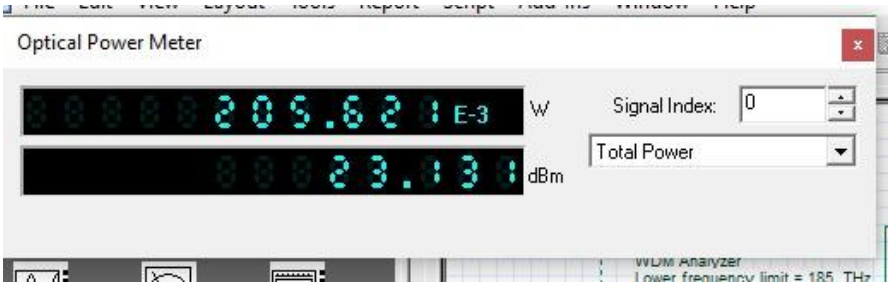
SOFTWARE REQUIRED: Opti System.

CIRCUIT DIAGRAM:



OUTPUTS:


Optical Spectrum Analyzer
Optical Spectrum Analyzer
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Wavelength (m)
Power (dBm)
Signal Index:
Auto Set
Wavelength
Amplitude
Resolution Bandwidth
Invert Colors
BER Analyzer
BER Analyzer
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Q Factor
Min BER
Threshold
Height
BER Pattern
Show Eye Diagram
Analysis
Max. Q Factor 4.88337
Min. BER 4.84975e-007
Eye Height 0.0270268
Threshold 0.0380388
Decision Inst. 0.523438
Invert Colors
Color Grade
Patterns
Calculate Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
BER Analyzer_1
Max. Q Factor 4.86909
Min. BER 5.1474e-007
Eye Height 0.0268445
Threshold 0.037942
Decision Inst. 0.523438
BER Analyzer_3
Max. Q Factor 4.76213
Min. BER 8.7257e-007
Eye Height 0.0256717
Threshold 0.0373099
Decision Inst. 0.523438


Optical Power Meter
205.621 E-3 W
23.131 dBm
Signal Index: 0
Total Power

REFLECTION:

learned how multiple optical channels are transmitted through a single fiber using DWDM.

1. I understood the function of an EDFA in compensating for fiber attenuation.
2. I learned to analyze BER and Q-factor using BER analyzers.
3. I understood how fiber length affects optical communication performance.

RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & DWDM Setup	Correct selection and connection of OptiSystem components for the DWDM link (PRBS Generator, NRZ Pulse Generator, CW Lasers, WDM Multiplexer, Optical Fiber, EDFA, WDM Demultiplexer, PIN Photodetectors, Low Pass Filters, BER Analyzers).	All four transmiLer channels, multiplexer, fiber, EDFA, demultiplexer and receiver chains are correctly selected, connected in sequence and configured without errors.	All components are present with minor connection or layout errors, but the DWDM link remains functional.	Component selection is partially complete; missing receiver chains or EDFA placement errors impede signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Execution	Accurate setup of input parameters: Transmit Power (20 dBm), Center Frequency (193.1 THz), Channel Spacing (100 GHz), Fiber Length (50 km, 75 km, 100 km) and Loop Control (0).	All input parameters and fiber-length test cases are accurately set and executed, with and without the EDFA.	Test case parameters are mostly accurate with minor seLing discrepancies.	Basic parameters are configured, but multiple fiber-length cases or the with/without EDFA comparison are missing.	Incorrect parameter seLings or failure to run the required test cases.	20%
Measurement & Performance Monitoring	Accurate extraction and recording of Min BER, Max QFactor, received power, optical spectrum and eye diagrams for all channels and fiber lengths.	Precise data extraction across all fiber lengths, neatly tabulated with complete BER, Qfactor and power readings and clear spectrum/eye diagrams.	Data is collected with minor omissions in BER, Q-factor or power readings.	Partial data collection with incomplete tables or missing spectrum/eye diagram captures.	Inaccurate, incomplete or missing data and measurements.	20%

Amplification & Fiber Loss Analysis	Detailed evaluation of fiber aLenuation, dispersion, EDFA gain and noise, and the effect of amplification on received power, Qfactor and BER.	Comprehensive analysis correctly relating fiber length, aLenuation and EDFA compensation to received power, Q-factor and BER, with and without amplification.	Clear analysis identifying basic trends in power, BER and Q-factor relative to fiber length and EDFA use.	Basic aLempt at result analysis without explaining physical mechanisms such as aLenuation, dispersion or amplifier gain.	Erroneous or missing analysis of simulation results.	20%
Report & Conclusion	Identification of the benefit of EDFA amplification, achievable transmission distance, trade-offs (amplifier noise, channel spacing), structure and clarity of the report.	Report features clear spectra and eye diagrams, complete result tables and insightful conclusions identifying optimal operating parameters for the DWDM link.	Satisfactory report with relevant displays and a sound concluding summary.	Incomplete report with missing graphs, readings or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

CONCLUSION :

The DWDM optical communication system was designed and simulated successfully using OptiSystem. The performance of the link was analyzed for different fiber lengths, and the role of the EDFA in compensating for fiber attenuation was studied. The BER and Q-factor results were recorded to evaluate signal quality. The experiment demonstrated the importance of optical amplification in long-distance, high-capacity optical communication systems.